

SCI RIGOR: EVALUATING OPEN-ENDED SCIENTIFIC ANALYSIS BEYOND FINAL SCORES

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ABSTRACT

Scientific coding agents increasingly perform long-horizon, open-ended workflows that produce interdependent code, numerical results, figures, and claims. Existing evaluations typically score workflow completion or final artifacts in isolation. These scores do not establish whether a reported claim is supported by the data, computation, and visualization produced in the same run. We formulate *evidence-grounded scientific analysis with multimodal artifacts* as an evaluation task. Given a scientific question, analysis-ready data, and study context, an agent must produce an executable analysis, structured results, a visualization, and atomic scientific claims. Success requires the required findings to be reported and every primary claim to retain a verified same-run support path. **SCI RIGOR** couples a fine-grained evaluation framework with an evidence-complete benchmark for this task. The framework reconstructs generated artifacts as a typed evidence graph, evaluates artifact fidelity separately from edge validity, scores claims over complete computational and visual paths, and localizes the earliest unsupported relation. Source-grounded alternative paths accommodate scientifically equivalent methods and visual encodings, while weakest-link scoring exposes upstream errors behind plausible downstream outputs. The benchmark comprises 100 cases drawn from scientific articles published in journals such as *Nature*, spanning six domains and 17 subfields. Each case aligns the question, complete inputs, executable source analysis, numerical results, source figure, atomic findings, and provenance. We evaluate 11 agent/model configurations on SCI RIGOR. On full-benchmark runs, agents’ claims agree with their own results at nearly the same rate whether those results are faithful to the scientific target or not (91.8% versus 91.0%). Yet no system exceeds 62.6% on the soft evidence-chain score or 18.0% strict whole-chain success. Internal coherence is therefore not sufficient evidence of scientific correctness: support must be verified along the full data-to-claim path.

1 INTRODUCTION

Scientific coding agents are moving beyond isolated code generation toward open-ended analysis workflows that produce numerical results, figures, and scientific conclusions (Chen et al., 2025; Wang et al., 2026a;b). Because these outputs may enter papers, reports, and downstream decisions, their evidential validity matters as much as their surface plausibility. These artifacts may look credible on their own while failing to support one another. Consider a case in our collection that asks how food-delivery workload changes under extreme heat (Appendix F.2). The source regression stores an estimate and its confidence limits on a log scale, so a correct analysis must apply $100[\exp(x) - 1]$ to all three values. If an agent transforms only the estimate, it can still state the correct-looking finding of an 18.0% increase (95% CI 12.6–23.6%), even though its structured interval and plotted uncertainty band remain on the wrong scale. The sentence matches the scientific target, but the submitted computation and figure do not support it. Evaluating such workflows therefore requires more than final-output scoring, which judges whether a returned artifact looks plausible or resembles a reference without verifying how it was produced.

Existing benchmarks score complementary parts of this workflow. Scientific programming benchmarks assess code, execution, and analysis decisions (Chen et al., 2025; Zhang et al., 2026; Gu et al., 2024); research-agent benchmarks extend evaluation to hypotheses, experiments, and multi-artifact

outputs (Majumder et al., 2025; Wang et al., 2026b; Xu et al., 2026); and visualization or claim-auditing benchmarks examine figures and scientific statements (Ai et al., 2026; Deng et al., 2026; Yue et al., 2026). Across these settings, the scored object is typically a program, stage, artifact, or final outcome. They therefore do not jointly establish whether a newly generated claim is supported by the data, computation, and visualization produced in the same run. This blind spot can accept a correct-sounding but unsupported conclusion, while exact-reference matching can reject a defensible alternative analysis.

We formulate *evidence-grounded scientific analysis with multimodal artifacts* as the post-experiment task between collected data and a reportable conclusion. Given a scientific question, analysis-ready data, and study context, an agent produces mutually supporting code, results, a figure, and atomic claims. It may choose among defensible methods and visual encodings. Task success requires the required findings to be reported and every primary claim to retain at least one complete computational and visual support path. The path comprises observable scientific artifacts and dependencies rather than hidden reasoning or proof steps. We ask: **RQ1:** How faithfully do agents construct each artifact? **RQ2:** Does agreement among an agent’s own artifacts imply agreement with the scientific target? **RQ3:** How often does local success survive composition into a complete data-to-claim path?

A suitable evaluation must ground every scored artifact in executed data, reconstruct relations among artifacts from runtime evidence, and accommodate scientifically equivalent methods and visual encodings. SCIRIGOR meets these requirements with a typed evidence graph. During benchmark construction, source code, data, results, figures, and findings define human-audited support paths. During evaluation, the evaluator reconstructs the candidate graph from executed artifacts, scores artifact fidelity separately from edge validity, assigns each claim the strength of its best pre-specified support path under a weakest-link rule, and localizes the earliest unsupported relation. Complete-path scoring exposes plausible claims whose computational or visual support has already failed, while source-grounded alternative paths admit valid non-reference analyses.

We instantiate this framework in 100 evidence-complete cases drawn from scientific articles published in journals such as *Nature*, spanning six domains and 17 subfields (Figure 2). Each case aligns a question with all code-consumed data, executable source analysis, numerical results, a source figure, atomic findings, and provenance. The agent receives only the question, declared inputs, and study context; every admitted case passes two clean reproductions and a final human audit. We evaluate 11 agent/model configurations on SCIRIGOR. On full-benchmark runs, agents’ claims agree with their own results at nearly identical rates whether those results are faithful to the scientific target or not (91.8% versus 91.0%), while the best strict whole-chain success rate is only 18.0%. These results show that internal coherence is not sufficient evidence of scientific correctness.

Our contributions are:

- **Task formulation.** We formulate evidence-grounded scientific analysis with multimodal artifacts. The task connects generated code, results, a figure, and atomic claims through verifiable same-run support (Section 3.1).
- **Evaluation framework.** We define the scored object as a verified data-to-claim support path, separate artifact fidelity from edge validity, and localize the earliest unsupported relation in a typed evidence graph (Section 3.2).
- **Benchmark.** We construct 100 evidence-complete cases spanning six domains and 17 subfields, with executable reconstruction, source-grounded alternative paths, and human-audited provenance (Section 4).
- **Empirical diagnosis.** We evaluate 11 agent/model configurations and quantify where execution, scientific fidelity, self-consistency, and end-to-end evidence composition diverge (Section 5).

2 RELATED WORK

Scientific-agent benchmarks differ in their primary scored object: an executable analysis, a completed research workflow, or an individual figure or claim. SCIRIGOR connects these levels by scoring a generated claim together with the computation and visualization produced to support it. Table 1 summarizes the most relevant comparisons.

Table 1: Support coverage of scientific-agent benchmarks.

Benchmark	Code	Figure	Generated claim	Claim path	First failure
ScienceAgentBench (Chen et al., 2025)	✓	△	✗	✗	✗
BLADE (Gu et al., 2024)	✓	✗	✗	△	✗
DiscoveryBench (Majumder et al., 2025)	△	✗	✓	✗	✗
DataSciBench (Zhang et al., 2026)	✓	△	△	✗	✗
FIRE-Bench (Wang et al., 2026b)	✓	△	✓	✗	△
SciVisAgentBench (Ai et al., 2026)	✓	✓	✗	✗	✗
SCIRIGOR	✓	✓	✓	✓	✓

Symbols: ✓ = required and directly evaluated; △ = subchain-, rubric-, contract-, or stage-level coverage; ✗ = outside the primary scope. *Claim path* requires verification of same-run computational and visual support; *First failure* localizes the earliest unsupported evidence edge within an instance.

Scientific programming and data-analysis benchmarks. Scientific programming benchmarks make executable code or analytical decisions the primary unit. NatureBench (Wang et al., 2026a) tests reproduction or improvement of published results in standardized environments, while ScienceAgentBench (Chen et al., 2025) scores programs and their execution on paper-derived tasks. BLADE (Gu et al., 2024) evaluates open-ended choices of variables, transformations, and statistical models, and DataSciBench (Zhang et al., 2026) evaluates practical data-science agents through executable tasks and artifact-specific metrics. These benchmarks verify whether an analysis is executable or scientifically defensible, but do not extend the scored unit through a generated figure to its reported claims.

Scientific discovery and end-to-end research agents. Research-agent benchmarks broaden the unit from one analysis to a multi-stage workflow. DiscoveryBench (Majumder et al., 2025) evaluates discoveries from supplied data; FIRE-Bench (Wang et al., 2026b) evaluates rediscovery from research questions through experiments and conclusions; and ResearchClawBench (Xu et al., 2026) scores complete research artifacts with multimodal rubrics. FrontierChallenge (Su et al., 2026) uses deliverable-specific executable graders for fixed-input workflows. These benchmarks evaluate whether a workflow delivers the required research outputs, whereas SCIRIGOR asks whether each generated claim retains a verified support path within that completed workflow. SciDataBench (Anonymous, 2026) further studies agents operating over scientific data APIs.

Visualization, claim verification, and evidence chains. Visualization and claim-verification benchmarks isolate individual links in the same chain. SciVisAgentBench (Ai et al., 2026) evaluates multi-step scientific analysis and visualization, while SciFigQual-Bench (Deng et al., 2026) scores existing figures against captions and manuscript context. FactReview (Yue et al., 2026) audits claims extracted from papers using literature and execution evidence. SCIRIGOR instead jointly evaluates newly generated results, figures, and claims, reconstructs their dependencies from runtime evidence, and localizes the earliest unsupported edge.

3 SCIRIGOR

SCIRIGOR evaluates scientific analysis as a connected evidentiary object. It defines what an agent must produce, reconstructs the dependencies among the executed artifacts, and scores whether each reported claim retains complete computational and visual support. This progression separates three questions that final-output scores conflate: whether an artifact is faithful to the scientific target, whether one artifact supports another, and whether those local relations compose into an end-to-end data-to-claim path.

3.1 TASK FORMULATION

We formulate *evidence-grounded scientific analysis with multimodal artifacts* as the post-experiment task between collected data and a reportable conclusion. Let q denote a scientific question, $\mathcal{D}$ the analysis-ready data, and κ the study context and measurement semantics. An agent $\mathcal{A}$ produces four

interdependent scientific artifacts:

$$\mathcal{A}(q, \mathcal{D}, \kappa) \mapsto \hat{\mathcal{Y}} = (\hat{T}, \hat{R}, \hat{V}, \hat{C}), \quad (1)$$

where $\hat{T}$ is an executable analysis, $\hat{R}$ its structured numerical results, $\hat{V}$ a data-grounded visualization, and $\hat{C}$ a set of atomic scientific claims. Multimodality refers to the generated visualizations and textual claims, whose support is traced through code and numerical results. The joint output makes the relations among artifacts part of the task rather than treating each artifact as an independent endpoint.

The task is executable: running $\hat{T}$ over $\mathcal{D}$ must regenerate $\hat{R}$, $\hat{V}$, and $\hat{C}$ and expose provenance $\hat{P}$ for evaluator verification. A successful submission covers the required findings and gives every primary claim at least one verified computational and visual support path through artifacts generated in the same run. Pre-specified, source-grounded alternatives represent scientifically valid choices of analysis method and visual encoding. Appendix A, Table 3, maps these abstract artifacts to the serialized submission contract.

3.2 EVIDENCE-GRAPH EVALUATION

SCI RIGOR constructs the reference support specification once, before model evaluation. Curators execute the frozen source workflow and canonicalize its consumed data views, transformations, statistical results, visual marks, and atomic findings into semantic slots. Each required claim slot is linked to at least one computational support set and one visual support set. Items within a set are jointly required, whereas separate sets represent approved alternative methods or visualizations. A final human audit verifies that the selected panel, source passage, author code, consumed data, claim scope, and support sets address the same scientific question. The resulting hidden specification determines the relations to verify without prescribing candidate filenames, identifiers, code, prose, or pixels.

At evaluation time, the submission becomes an executed dependency graph over observable scientific artifacts rather than an agent-authored reasoning trace. The evaluator reconstructs this graph; the agent does not submit it as an additional artifact. Its core relations are

$$D \rightarrow T \rightarrow R \rightarrow C, \quad T \rightarrow V, R \rightarrow V, V \rightarrow C, \quad (2)$$

where D , T , R , V , and C denote data assets, transformations, statistical results, visual marks, and atomic claims. The evaluator instantiates D and T from runtime traces, R from regenerated candidate results and independent numerical oracles, V from captured plotting objects and arrays, and C from structured claim records. Here the unhatted symbols denote artifact types, whereas the hatted symbols in Equation 1 denote a candidate’s realized outputs. Candidate-declared provenance and support pointers propose relations, which the evaluator verifies against these executed observations. Primary claims require both a complete computational path and claim-relevant visual support with matching population, grouping, quantity, direction, and required uncertainty.

3.3 ARTIFACT AND RELATION VERIFICATION

The evaluator scores graph nodes and edges separately so that an incorrect artifact can be distinguished from an invalid dependency between otherwise plausible artifacts. Candidate artifacts are canonicalized and matched to hidden semantic slots under type and scope constraints. Transform fidelity checks data views, selectors, operation families, and parameters. Result fidelity checks the estimand, analysis unit, method family, numerical value, and uncertainty against independent recomputation. Visual fidelity checks bound arrays, encodings, groups, units, scales, and uncertainty, while claim fidelity checks direction, magnitude, population scope, inferential strength, and causal ceiling.

Edge verification follows the same executed evidence. The evaluator tests whether the observed data produced the transformation, the transformation produced the result, the rendered marks encode the same quantities, and the result and marks support the claim. Appendix A, Table 4, specifies the evidence and pass condition for every relation. Lineage, numerical recomputation, visual binding, and graph connectivity are checked programmatically. A frozen semantic matcher handles residual paraphrase and scope matching for atomic claims but cannot override a failed structured, numerical, or provenance check.

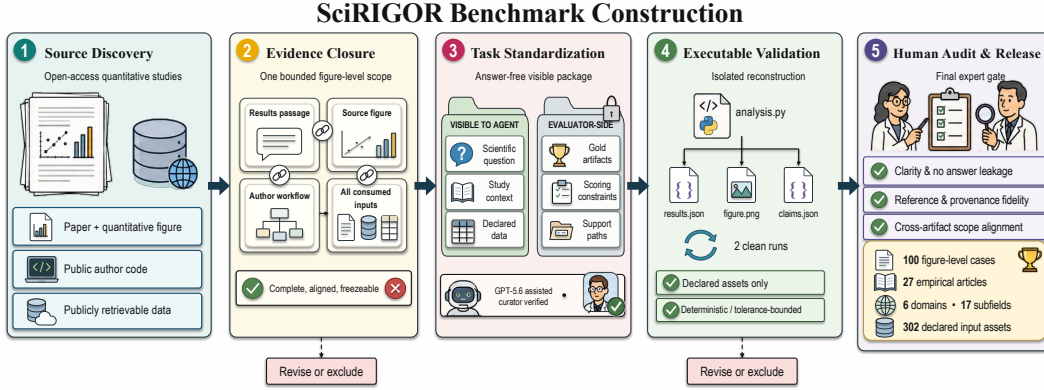


Figure 1: **SCI-RIGOR benchmark construction.** Open-access studies enter the benchmark only after figure-level evidence closure, task standardization, two clean reconstructions, and expert audit. The agent-visible task is separated from evaluator-side gold artifacts, scoring constraints, and support paths.

3.4 CLAIM-PATH SCORING AND FAILURE LOCALIZATION

The hidden specification represents accepted method families, visual encodings, and claim-support rules as semantic constraints rather than a single gold program or image. Let $\mathcal{P}_{\text{comp}}(c)$ and $\mathcal{P}_{\text{vis}}(c)$ denote the accepted computational and visual support paths for claim c , and let each verified node or edge item z have score $s(z) \in [0, 1]$. SCI-RIGOR computes

$$S_k(c) = \max_{p \in \mathcal{P}_k(c)} \min_{z \in p} s(z), \quad k \in \{\text{comp}, \text{vis}\}, \quad S(c) = \min\{S_{\text{comp}}(c), S_{\text{vis}}(c), F_C(c)\}, \quad (3)$$

where $F_C(c)$ is local claim fidelity. The inner minimum applies the weakest-link rule to one path, and the outer maximum admits any complete, approved alternative. Generated claims are matched one-to-one to required claim slots, and their verified support mass yields evidence-chain precision, recall, and F1. Strict evidence-chain success requires execution, complete support for every required claim, no unsupported primary claim, and satisfaction of all hard visual constraints. Artifact fidelities, edge pass rates, and an execution-gated soft chain aggregate provide complementary diagnostic views.

For an incomplete chain, SCI-RIGOR reports the earliest topological layer from which no accepted partial path can reach the claim: contract/execution, Data→Transform, Transform→Result, Transform/Result→Visual, or Result/Visual→Claim. An incomplete path does not determine the score when another independently verified path remains complete, while downstream presentation quality cannot repair an upstream scientific error.

4 SCI-RIGOR BENCHMARK

The SCI-RIGOR benchmark instantiates the evaluation framework with 100 evidence-complete, figure-level cases from 27 empirical articles. Each case aligns a scientific question with all code-consumed inputs, an executable source analysis, numerical results, a source visualization, atomic findings, and provenance within one experimental scope. We construct this alignment through source discovery, evidence closure, standardized task assembly, executable reproduction, and final human audit (Fig. 1).

4.1 SOURCE DISCOVERY AND EVIDENCE CLOSURE

We searched open-access quantitative studies for figures accompanied by public source data and author-released plotting code. Discovery combined a Europe PMC search of the *Nature* and *Science* main journals with audited Nature Portfolio and other open-access sources. For each candidate, automated indexing connected in-text figure references with full-text passages, supplements, Data

Availability statements, repository records, and exposed source-data links. These links formed a review queue; admission depended on evidence completeness rather than publication venue (Appendix B).

An admitted case contains four mutually aligned source components: an author-generated quantitative figure from an auditable paper version; a bounded Results passage that describes and interprets the selected figure or panel; the complete author-released script or notebook that generates that target; and every raw, processed, or intermediate file consumed by its entry point. The plotting program must have a stable source locator, and every data dependency must be publicly retrievable and freezeable. Together, these requirements tie the visual target, experimental interpretation, executable analysis, and inputs to the same scientific question.

Evidence closure required two clean executions of the frozen author entry point against the complete inputs. The reproduced quantities, visual marks, groups, units, and uncertainty encodings had to agree with the paper figure and its analysis. Each accepted case covers one figure or one coherent panel subset, and its question, passage, code, inputs, and findings are bounded to that scope. The visual reference is an original source raster or an audited source-pixel crop. Missing dependencies, unrecoverable semantics, or substantive disagreement led to exclusion under the rules detailed in Appendix C.

4.2 TASK CONSTRUCTION AND STANDARDIZATION

Each closed source bundle became a self-contained task without answer-bearing content. We retained an explicit paper question when it matched the selected figure scope; otherwise, curators wrote a neutral question bounded to the experiment. The visible package contains the question and study context in the task specification together with the declared analysis-ready data. Source references, gold artifacts, construction metadata, and scoring rules remain evaluator-side. Appendix D specifies the question, visibility, storage, and serialized interface.

Standardization began after source closure. GPT-5.6 assisted with locating target-specific operations, drafting neutral task descriptions, mapping heterogeneous tables into documented assets, and proposing source-grounded Python translations and structured records. Curators verified every proposal against the frozen author workflow and retained unchanged author code for audit. Original Python operations were preserved; non-Python workflows were translated into Python ports that retained source selectors, transformations, estimators, scales, uncertainty layers, and plot structure. Execution and visual-semantic comparison validated each port before case admission.

4.3 EXECUTABLE QUALITY CONTROL

Every assembled case ran twice in an isolated environment containing only its executable reference analysis and declared inputs. Both runs had to terminate successfully, regenerate the required results, visualization, and claims, support provenance reconstruction, consume only declared assets, and agree under a deterministic or explicitly tolerance-bounded contract. Structured hidden constraints were then checked against the regenerated numerical results, visual semantics, data bindings, and atomic-claim support. All 100 released cases passed both clean reconstruction runs.

Curators completed a final audit of question clarity, answer leakage, reference fidelity, scoring-constraint coherence, and alignment among the selected panel, source passage, author code, consumed data, and claims. Unresolved scope, provenance, or support discrepancies triggered revision or exclusion. This construction-time audit fixes the hidden reference and support rules, enabling routine model submissions to be scored automatically. Candidate methods outside the frozen alternatives enter expert adjudication; accepted alternatives are added to the support specification before rescoring. Appendix E provides the complete execution, asset, and review protocol.

4.4 BENCHMARK COMPOSITION

The release covers six reporting domains defined by the primary scientific subject of each source study: economics and policy (24 cases); computing and information sciences (23); social and behavioral sciences (22); cognitive science (15); life and health sciences (9); and physical sciences (7).

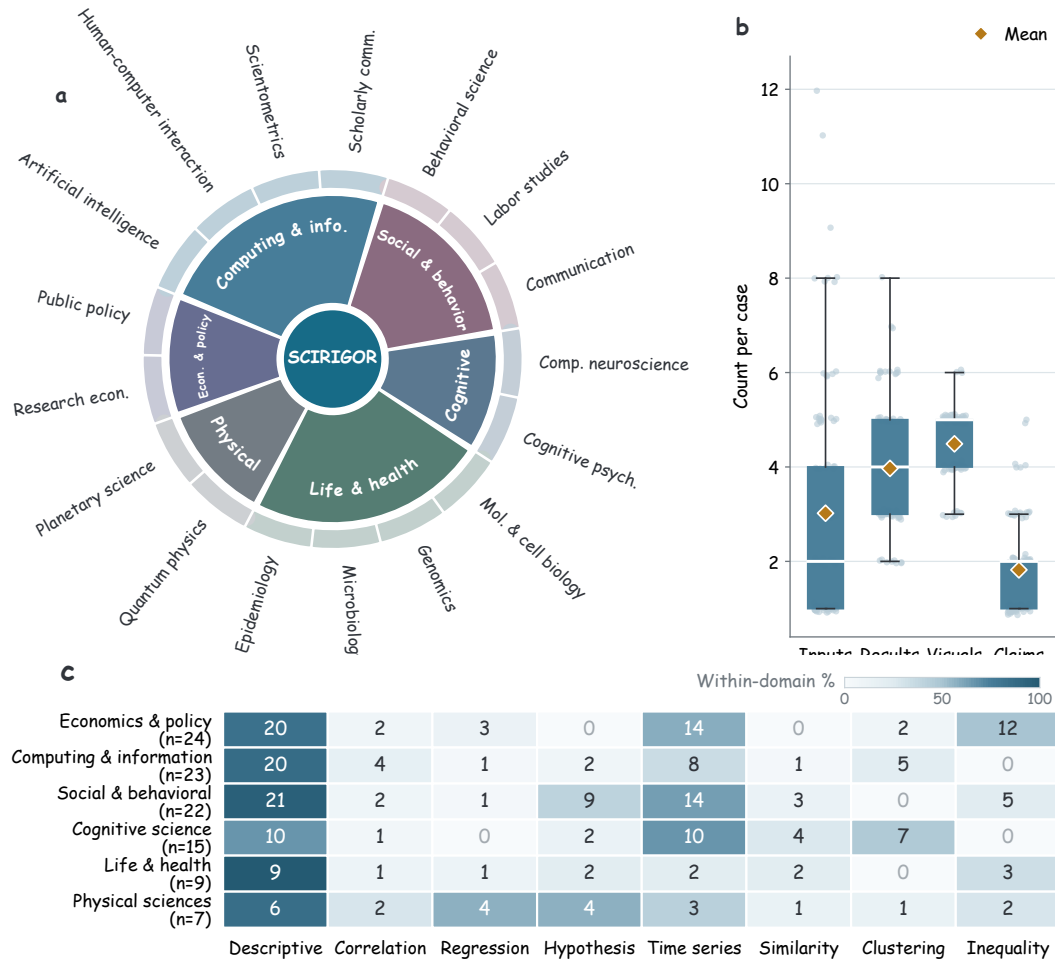


Figure 2: SCIRIGOR spans diverse scientific domains, analysis methods, and multi-artifact task complexities. (a) The 100 cases span six reporting domains and 17 disciplinary subfields. The taxonomy ring assigns equal angular width to every subfield, while each inner domain spans the subfields it contains. (b) Cases require multiple input assets, result slots, visual requirements, and atomic claims, creating long evidence chains. (c) Method families are multi-label across domains; cells report case counts, and darker color denotes greater within-domain prevalence.

Multiple cases may originate from one article because the benchmark unit is a bounded figure-level question. The released metadata records the complete case-to-subfield mapping shown in Fig. 2a.

The cases vary in the number of input assets, required result slots, visual requirements, and atomic claims, producing evidence chains of different lengths (Fig. 2b). Analysis methods are multi-label: descriptive aggregation appears across all domains, physical sciences contain more regression, model-fitting, and inferential checks, cognitive science contributes trajectory, similarity, and classification analyses, and economics/policy contains many distribution and inequality computations (Fig. 2c). These distributions describe coverage rather than model performance. They reflect the availability of figure-specific public code and complete code-consumed data, not a probability-balanced sample of the scientific literature.

Table 2: Main results on SCI-RIGOR (%; higher is better). Blue shading is normalized within each model row.

Model	Exec.	T-Fid.	R-Fid.	V-Fid.	C-Fid.	R→C	SoftECS	Strict
GPT-5.5 (xhigh)	100.0	57.9	68.2	72.8	57.2	93.4	62.6	18.0
GPT-5.5 (high)	100.0	58.5	69.8	70.8	59.1	91.3	59.2	16.0
GPT-5.6-Sol (high)	100.0	59.5	64.7	72.8	58.7	90.0	61.1	13.0
GPT-5.6-Sol (xhigh)	100.0	58.7	64.8	70.9	57.7	88.9	57.6	11.0
GPT-5.5 (low)	63.0	58.3	50.4	75.4	46.4	97.8	28.0	3.0
Qwen3.8-Max	47.0	24.1	18.2	34.0	23.5	46.5	20.1	2.0
Kimi-K3	79.0	43.5	36.4	55.9	40.2	73.2	35.0	1.0
GPT-5.6 (low)	88.0	55.5	47.2	77.3	52.5	97.9	43.0	0.0
Claude Sonnet 5	75.0	44.5	40.6	54.3	24.0	71.9	22.1	0.0
Qwen3-Coder-30B-A3B	29.0	9.3	7.8	17.6	5.0	21.1	1.7	0.0
Qwen-AgentWorld-35B-A3B	12.0	5.1	3.4	7.1	6.1	11.4	0.0	0.0

5 EXPERIMENTS

5.1 MAIN RESULTS

Table 2 evaluates 11 agent/model configurations under the five-artifact interface. Metrics are percentages over each run’s evaluated cases; execution failures and missing required artifacts receive zero. We report artifact fidelity, Result→Claim consistency, and evidence-chain composition to distinguish local correctness from end-to-end scientific support.

Execution remains a substantive bottleneck. Execution gates every downstream scientific score. Kimi-K3 executes 79.0% of its assigned cases, Qwen3.8-Max executes 47.0%, Qwen3-Coder-30B-A3B reaches 29.0%, and Qwen-AgentWorld-35B-A3B reaches 12.0%. Scores computed only over successful outputs would therefore conflate scientific quality with selective completion.

Candidate self-consistency can mask scientific error. Across all 11 rows, R→C is the highest or second-highest displayed score, while result and claim fidelity are substantially lower for several high-execution runs. GPT-5.6 (low), for example, obtains 97.9% Result→Claim consistency but only 47.2% result fidelity and 52.5% claim fidelity. Its claims therefore tend to agree with its own reported results even when those results depart from the scientific target. Candidate self-consistency is necessary for a coherent submission but is not sufficient evidence of scientific correctness.

Partial success collapses under chain composition. The strongest run reaches 62.6% SoftECS but only 18.0% Strict success. Kimi-K3 falls from 79.0% execution to 35.0% SoftECS and 1.0% Strict success. These gaps show that disconnected artifact averages are insufficient: a submission can accumulate several locally successful components without completing a single fully supported data-to-claim path. Fidelity and edge consistency should be reported separately before applying an end-to-end chain gate.

5.2 ERROR ANALYSIS

We diagnose failures from three complementary views: the earliest broken stage, the full set of artifact and edge errors after successful execution, and whether local consistency distinguishes faithful from unfaithful results. These labels are produced by the evaluator and may overlap in Fig. 3b; they describe observable failure signatures rather than assigning a unique causal explanation to a trajectory.

Most chains break before a faithful result is established. Among 936 non-strict trajectories, 48.2% first fail in data selection or transformation and 30.1% fail at the artifact contract or execution gate (Fig. 3a). Result computation accounts for a further 11.9%, whereas visualization, claims, and provenance together account for only 9.8% of first failures. Thus, late-stage errors are visible, but the dominant bottleneck is establishing a valid computational basis for downstream evidence.

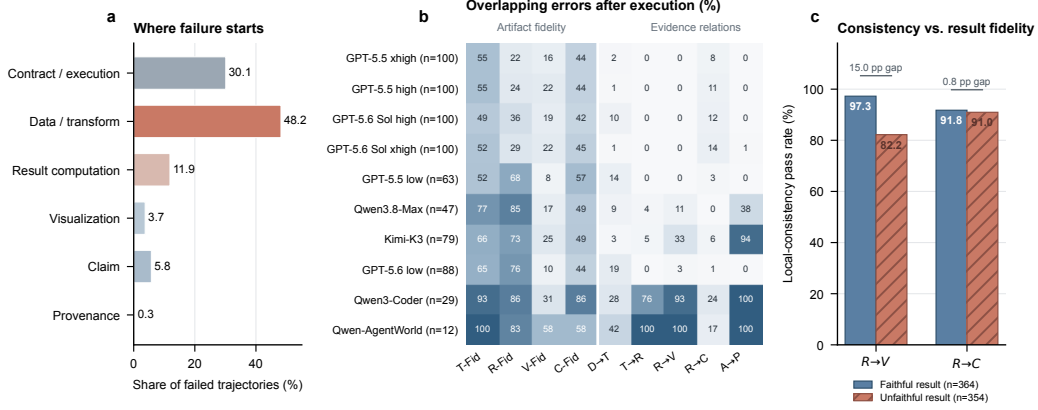


Figure 3: Case-level error analysis across ten complete 100-case runs. (a) Earliest failure stage among the 936 trajectories that do not complete a strict chain. Each trajectory contributes to exactly one stage. (b) Overlapping artifact- and edge-level errors among the 718 successfully executed trajectories; values are percentages within each model’s executed cases. (c) Local Result→Visual and Result→Claim consistency conditioned on independent result fidelity.

Scientific errors persist after execution succeeds. Conditioning on the 718 executed trajectories removes mechanical non-completion from the comparison. Transform fidelity fails in 59.6% of these trajectories, result fidelity in 49.3%, and claim fidelity in 47.9% (Fig. 3b). By contrast, Result→Claim consistency fails in only 8.6%. The model-level heatmap shows that this gap is not confined to a single run: systems frequently propagate their own intermediate values coherently even when those values do not match the scientific target.

Local self-consistency is nearly invariant to result correctness. Result→Claim consistency passes on 91.8% of result-faithful trajectories and 91.0% of result-unfaithful trajectories (Fig. 3c). Result→Visual consistency is more sensitive, but still passes on 82.2% of trajectories with unfaithful results, compared with 97.3% when results are faithful. A candidate can therefore explain and visualize its own result without establishing that the result is correct. This case-level conditional analysis confirms that cross-artifact coherence is necessary but cannot replace independent fidelity checks.

5.3 COMPLEXITY SENSITIVITY

Fig. 4 shows that complexity primarily affects chain completeness rather than surface claim plausibility. Moving from one input asset to at least five lowers Strict RG-EC from 45.7% to 25.0% and RG-EC F1 from 70.9% to 57.8%, whereas the gated-claim score drops from 71.7% to 63.7%. Required result slots show the same pattern: Strict RG-EC decreases from 43.9% to 30.8% and RG-EC F1 from 68.4% to 55.7%, while the gated-claim score remains nearly unchanged. This divergence supports scoring result-grounded claim paths rather than final claims alone. The highest visual-requirement bin contains only three cases and is therefore a stress-test signal, not a standalone estimate.

5.4 EVALUATION SCOPE

The experiments score observable artifacts submitted through the five-file interface. The evaluator reconstructs their semantic relations but does not observe latent reasoning or agent action trajectories. The results therefore compare artifact fidelity and complete evidence paths and localize observable breaks; they do not identify whether a failure arose from planning, implementation, or unsupported generation.

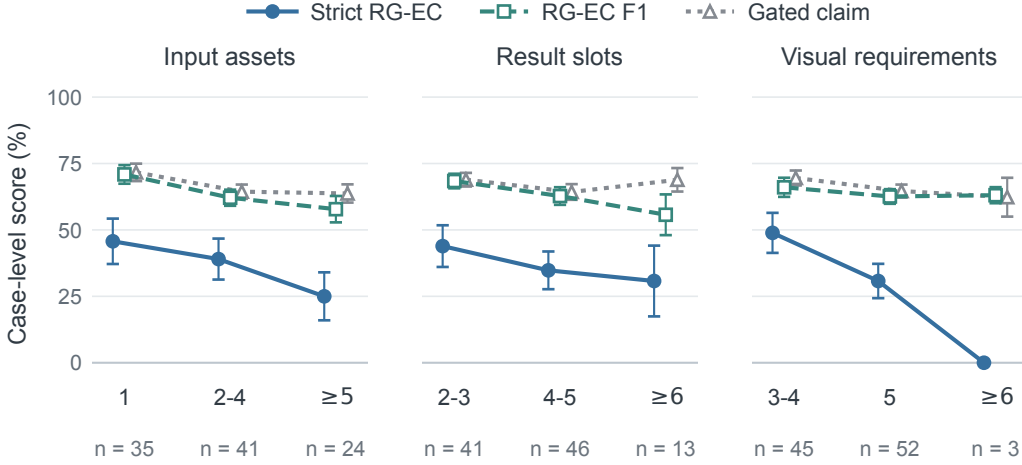


Figure 4: Performance by task complexity for GPT-5.6 Sol xhigh on 100 cases. Points show bin means; error bars denote ± 1 SEM across cases. Strict RG-EC decreases with more input assets, required result slots, and visual requirements; RG-EC F1 also declines with assets and result slots, while the gated-claim composite is flatter. RG-EC F1 gates claim coverage by verified numerical support, whereas Strict RG-EC requires complete support for every required and generated claim.

6 CONCLUSION

SCIRIGOR formulates evidence-grounded scientific analysis with multimodal artifacts around a verified data-to-claim support path. Its evaluation framework separates artifact fidelity from cross-artifact consistency, reconstructs computational and visual dependencies, accepts source-grounded alternative paths, and localizes the earliest unsupported relation. The benchmark’s 100 evidence-complete, figure-level cases align questions, inputs, source analyses, results, visualizations, findings, and provenance within one audited scope. Across 11 agent/model configurations, high execution or Result→Claim consistency can coexist with much lower target fidelity, and strict chain completion remains rare. Across the ten complete 100-case runs, 78.3% of failed trajectories first break at the contract/execution or data/transformation stage; among executed trajectories, Result→Claim consistency passes at nearly the same rate for faithful and unfaithful results (91.8% versus 91.0%).

Together, these results support the paper’s central evaluation principle: execution, visual plausibility, final-output similarity, and candidate self-consistency are not substitutes for independently verified relations from supplied data to each reported claim.

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Table 3: Five-file SCIRIGOR contract for reference and candidate bundles.

Serialized artifact	Frozen reference role	Candidate execution role
<code>analysis.py</code> (T)	Audited executable reconstruction of the gold analysis.	Model-written analysis and plotting entry point.
<code>results.json</code> (R)	Recomputed gold numerical results.	Model-computed numerical results.
<code>figure.png</code> (V)	Original paper figure or bounded source-pixel crop.	Data-generated visualization.
<code>claims.json</code> (C)	Data-verifiable atomic findings from the source analysis.	Model-generated atomic findings.
<code>minimal_provenance.json</code> (P)	Gold source and code-location hints.	Evaluator-verified input, execution, and code-location provenance.

Table 4: Independent evidence used to verify each candidate relation. Relation symbols follow Equation 2; both incident nodes must satisfy their semantic constraints before an edge can pass.

Relation	Evaluator evidence	Pass condition
$D \rightarrow T$	Sandbox file-access trace, selected row/column identifiers, operation parameters, and input/output fingerprints	The executed semantic data view and transformation agree with an allowed source view and operation family.
$T \rightarrow R$	Captured transform output and an independent case-specific numerical oracle	The estimand, analysis unit, method family, value, and required uncertainty recompute within the declared tolerance.
$T/R \rightarrow V$	Captured plotting arrays, artist bindings, and reconstructed Figure IR	Visual marks encode the executed quantities with the required population, grouping, channels, units, scales, and uncertainty.
$R/V \rightarrow C$	Structured claim fields, verified results and marks, and the hidden claim envelope	Direction, magnitude, population, uncertainty, inferential strength, and causal scope are entailed by at least one approved computational and visual path.

A EXECUTABLE CONTRACT AND EDGE-VERIFICATION DETAILS

The main paper presents the task interface and evidence-graph workflow at a conceptual level. Tables 3 and 4 provide the complete artifact ownership and programmatic edge-verification rules used by the evaluator.

B SOURCE DISCOVERY AND FIGURE-LEVEL INDEXING DETAILS

Search queries and source verification. One discovery stream queried the *Nature* and *Science* main journals through Europe PMC, combining journal ISSNs with open-access, full-text, research-article, and CC BY filters. We rechecked returned records for journal identity, article type, license, and source version. The ISSN constraint prevented expansion to other journals in the same publisher family. This crawler supplied candidates rather than defining venue eligibility: audited Nature Portfolio and other open-access sources faced the same evidence-completeness criteria.

Figure-level indexing. For each retrieved article, the pipeline collected full-text XML, source-native figures, machine-readable supplements, and exposed source-data links. It resolved in-text figure references and indexed the associated body paragraphs, quantitative statements, Methods sections, Data Availability statements, and repository identifiers. These signals constructed a review queue only; a detected figure–data link did not establish availability of target plotting code or all code-consumed inputs.

C EVIDENCE-COMPLETE CURATION DETAILS

Component-level exclusions. Schematics, decorative illustrations, and image-only panels without recoverable data bindings did not satisfy the figure requirement. Captions, headings, and nearby discussion of other experiments could not replace a bounded Results passage. General software, upstream preprocessing alone, code for another panel, and curator-written plotting code could not replace the original target plotting program, which required a stable locator such as a commit, file, function, line range, or notebook cell. A publisher summary table was insufficient when the target code also consumed unavailable raw data, mappings, model outputs, checkpoints, or configurations.

Reproduction audit. Reviewers ran the frozen author entry point twice in a pinned environment against complete frozen inputs and compared the reproduced quantities, marks, groups, units, and uncertainty encodings with the source figure and analysis. Private-path dependencies, unrecoverable semantics, or substantive disagreement with the published result caused rejection. The source registry retains the original source, environment, file-open traces, execution logs, and comparison report. Independent reconstruction could document a rejection but could not substitute for missing author code.

Panel and crop audit. We retained the full paper and source figure. When a case covered only part of a composite figure, its visual reference was a logged, unresized crop of the selected source panels. Recorded crop coordinates, dimensions, and hashes preserve the link to the source image; redraws were not accepted as references.

D STANDARDIZED CASE CONSTRUCTION DETAILS

Question checks and visibility boundaries. Curators checked every question for leading language, answer-bearing values, vague comparison sets, and undefined analysis units. The visible package contains only `input/task.json`, declared assets under `input/data/`, and the scientific context and measurement semantics needed to solve the task. The source paper, unchanged author code, original figure, gold findings, scoring rules, and construction metadata remain evaluator-side.

Evaluator-side storage. The `source_registry/` retains the source PDF, unchanged author code, data hierarchy, reproduction output, crop record, and execution logs; `hidden_evaluation/` stores source views and scoring constraints. Candidate figures are checked for marks, field bindings, axes, units, uncertainty channels, and panel semantics rather than pixel identity with the frozen paper raster.

GPT-5.6-assisted curation. Where applicable, GPT-5.6 helped locate target-specific operations, draft neutral task descriptions, map heterogeneous tables into documented visible assets, and propose source-grounded Python translations and structured records. Curators checked every proposal against the frozen author workflow. Only source-present code, data, figures, and claims could enter an accepted case.

Python-port validation. Original Python operations were retained. R, R Markdown, and other workflows were translated while preserving source selectors, transformations, estimators, scales, uncertainty layers, and plot structure. Execution and visual-semantic comparison validated each port, with library- or platform-sensitive differences recorded explicitly. Every `analysis.py` used only visible inputs, made no network calls, and regenerated `results.json`, `figure.png`, and `claims.json` without copying frozen references; the evaluator verified provenance separately.

E VALIDATION AND RELEASE DETAILS

Asset integrity and execution environment. The release contains 302 declared input assets, each materialized at its specified path, matched to its recorded SHA-256 hash, and stored as a regular file rather than a symbolic link. File-open traces confirmed that execution consumed only declared

Table 5: Disciplinary composition of the 100 SCIRIGOR cases. Domains are used for descriptive reporting and do not imply a probability-balanced sample of the scientific literature.

Reporting domain	Included research topics	Cases
Economics and policy	Research funding, award costs, funding inequality, peer review, scientific careers, and research-workforce composition	24
Computing and information sciences	Language-model reliability, human-computer interaction, preprint adoption, open-access pathways, bibliometrics, and scholarly communication	23
Social and behavioral sciences	Media representation, misinformation, platform labor, urban heat responses, collective behavior, and resource-allocation experiments	22
Cognitive science	Computational cognitive models, hippocampal learning, recognition, categorization, and human behavioral experiments	15
Life and health sciences	Cellular and molecular biology, genetics, microbiology, cancer biology, and influenza surveillance	9
Physical sciences	Atom interferometry, quantum sensing, parity readout, and post-impact asteroid photometry	7
Total		100

inputs. One numerically sensitive case used its pinned SciPy environment; all 100 cases passed two clean reconstruction runs.

Final human-review gate. Final curator approval was required for release. The audit assessed the visible task for ambiguity and answer leakage, verified a shared scientific scope across the selected panel, experimental passage, author code, consumed data, reference artifacts, and atomic claims, and checked hidden constraints for unsupported or contradictory requirements. Unresolved discrepancies triggered revision or rejection. All 100 released cases record approved review status.

Disciplinary assignment. Each case belongs to one reporting domain defined by the primary scientific system or institutional setting in the source paper, not by statistical method or visualization type. Table 5 summarizes the resulting composition.

F CASE STUDIES

We present six representative cases spanning the benchmark’s scientific coverage. Following the compact format of an illustrative task case, each example specifies the scientific task and frozen evidence, then summarizes the corresponding executable evidence chain. The displayed images are frozen paper figures or source-pixel panel crops used by the evaluator rather than candidate-generated visualizations. They illustrate the public benchmark contract and are not drawn from the hidden evaluation set.

F.1 ECONOMICS AND POLICY: NIH FUNDING INEQUALITY

Task and frozen evidence. This case asks how much of the inequality in NIH Research Project Grant funding is attributable to within-group and between-group differences when principal investigators are grouped by career stage, gender, race, or degree. The visible input contains the complete author data and four annual decomposition tables spanning fiscal years 1985–2020. Each table contains the total Theil index and its within- and between-group components, giving 144 group-year rows and 432 displayed point instances.

Reference evidence chain. The executable analysis verifies, for every year and grouping, that the two components sum to the total Theil index, preserves the author’s year and group selectors, and reconstructs the four-panel trajectory. As shown in Figure 5, within-group differences contribute more to funding inequality than between-group differences for all four groupings. This source-grounded finding is an accounting decomposition for funded RPG principal investigators, not a causal effect or a statement about applicants or the wider scientific workforce.

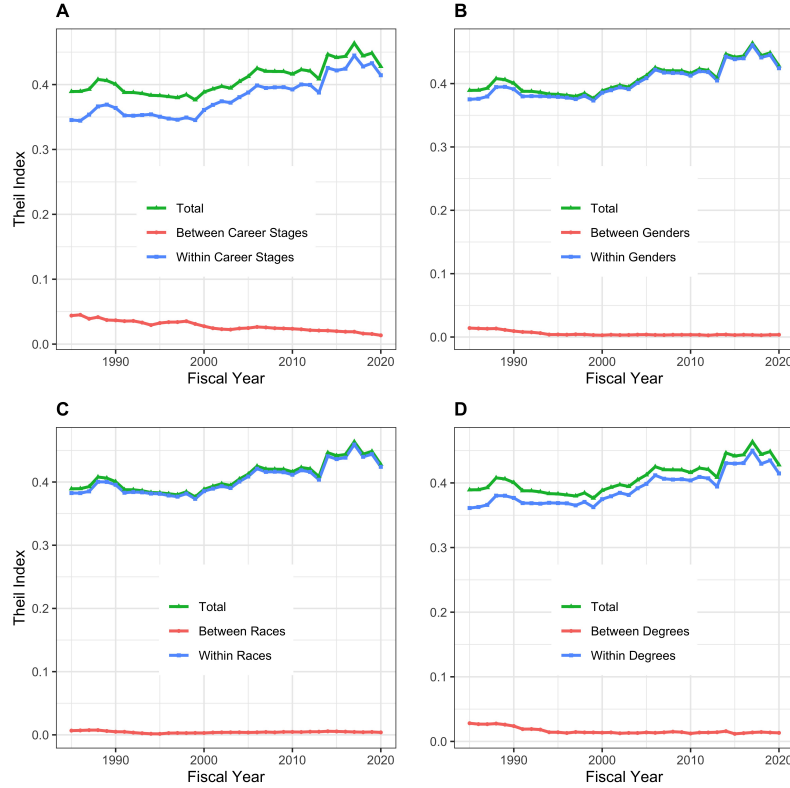


Figure 5: **NIH funding-inequality case.** The frozen Figure 4 reference decomposes total Theil inequality into between- and within-group components by career stage, gender, race, and degree from FY1985 to FY2020. The within-group component exceeds the between-group component throughout all four panels.

F.2 SOCIAL AND BEHAVIORAL SCIENCES: HEAT AND FOOD-DELIVERY WORKLOAD

Task and frozen evidence. This case examines how completed orders per food-delivery worker vary with daily maximum temperature, with particular attention to the lunch peak above 35°C . The three visible author tables contain processed regression outputs for all-day, lunch-peak, and dinner-peak workload across 21 temperature bins. Coefficients and both confidence bounds are on the log scale, so each must be transformed independently as $100[\exp(x) - 1]$ relative to the 20°C baseline.

Reference evidence chain. The normalized executable retains the author’s term ordering, applies the nonlinear transformation to all 63 estimates and confidence limits, and reconstructs the response curves and uncertainty bands. As shown in Figure 6, the bounded source analysis reports that lunch-peak workload is 7.3% higher in the $(35, 36]^{\circ}\text{C}$ bin (95% CI 4.8–9.7%) and 18.0% higher in the open-ended $(40, \infty)^{\circ}\text{C}$ bin (95% CI 12.6–23.6%). Because the inputs are processed model outputs rather than worker-level observations, the case supports these reported associations but no new causal claim.

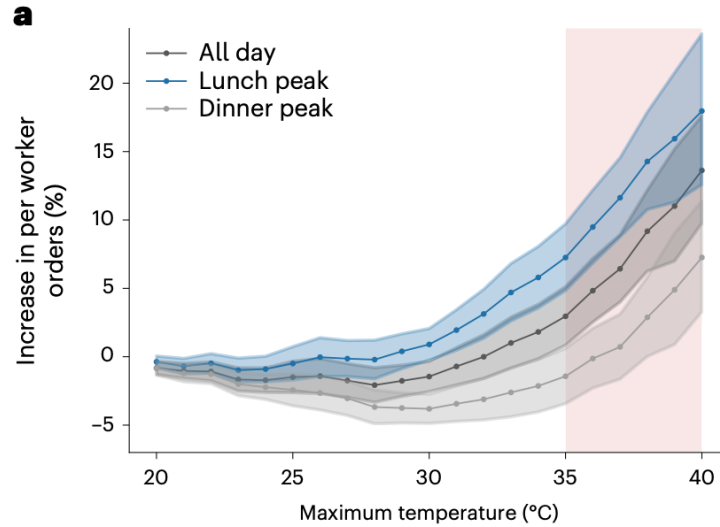


Figure 6: **Food-delivery workload case.** The frozen Figure 4a panel shows percentage changes in completed orders per worker relative to the 20 °C baseline for all-day, lunch-peak, and dinner-peak periods. Curves and bands encode transformed estimates and 95% confidence intervals; shading marks the extreme-heat range.

F.3 COMPUTING AND INFORMATION SCIENCES: LANGUAGE-MODEL RELIABILITY

Task and frozen evidence. This case asks how raw and shaped-up GPT, LLaMA, and BLOOM systems differ in correctness, robustness to prompt variation, concordance with human difficulty, and error-free behavior. Three author-released tables provide six normalized indicators for 32 named systems: 10 GPT, 10 LLaMA, and 12 BLOOM variants. The family sizes and raw-to-shaped composition are unequal, so pooling all systems would allow the largest family to dominate the comparison.

Reference evidence chain. The executable analysis first computes raw-versus-shaped summaries within each family and then averages the three family contrasts with equal family weight. It also reconstructs the common-scale radar panels from the complete system profiles. Figure 7 visualizes the source finding: shaped systems are more correct and more stable to prompt variation, but their correctness is less concordant with human difficulty and they exhibit more overall failures. The comparison is descriptive for this fixed system set; the aggregate tables contain no item-level replication or uncertainty with which to identify a causal effect of post-training.

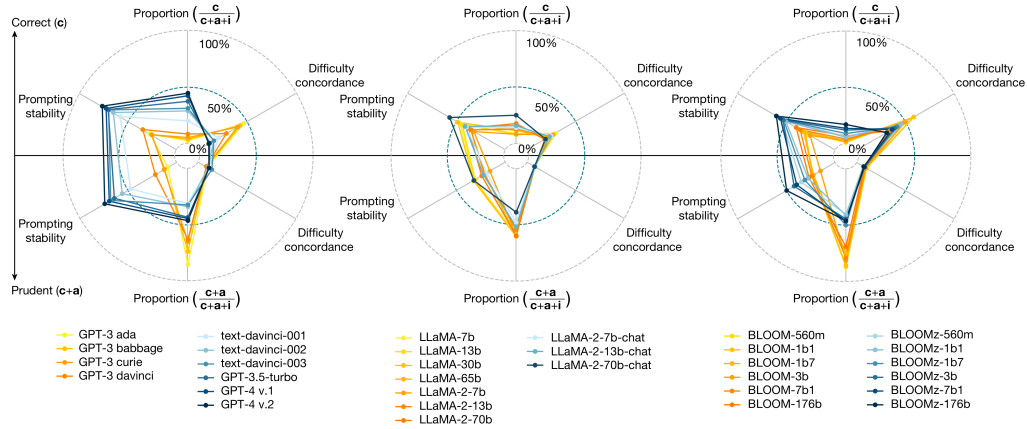


Figure 7: **Language-model reliability case.** The frozen Figure 1 reference presents six-indicator profiles for GPT, LLaMA, and BLOOM systems on a common radial scale. Warm profiles denote raw systems and blue profiles shaped systems, exposing the joint correctness, robustness, difficulty-concordance, and error-free trade-offs evaluated by the case.

F.4 COMPUTING AND INFORMATION SCIENCES: BIORXIV PUBLICATION OUTCOMES

Task and frozen evidence. This case asks what share of bioRxiv preprints was linked to a journal publication, which subject categories led by publication rate and count, and how the observed rate varied with posting date. The two complete author tables contain 61 monthly cohorts from November 2013 through November 2018 and 27 subject categories. Together they account for 37,648 preprints and 15,797 linked publications, with no imputation, smoothing, or model fitting required.

Reference evidence chain. The executable analysis recomputes every row-level proportion, aggregates the snapshot totals, and independently ranks categories by proportion and count. As summarized in Figure 8, the paper reports an overall linked-publication rate of 42.0%, the highest proportional rate for evolutionary biology (51.5%), and the largest linked count for neuroscience (2,608). It also describes rates near zero for the most recent preprints followed by an increase over roughly 12–18 months. The last pattern is explicitly snapshot-bound: recent cohorts have shorter follow-up, so their observed rate is not their eventual publication probability.

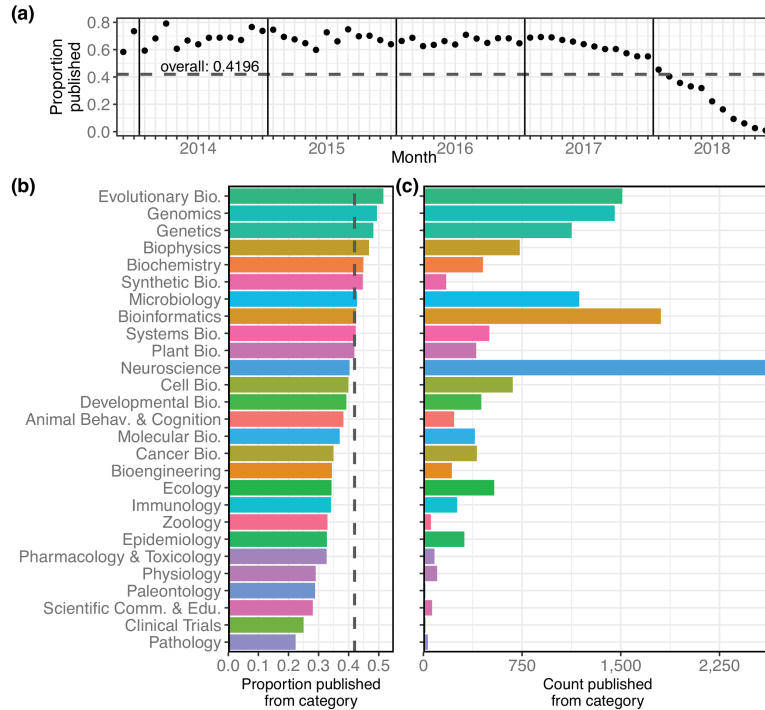


Figure 8: **bioRxiv publication-outcomes case.** The frozen Figure 3 reference combines monthly linked-publication proportions with category- level proportions and counts. Evolutionary biology has the highest linked proportion, whereas neuroscience has the largest linked count; the monthly view must be interpreted under unequal follow-up across posting cohorts.

F.5 LIFE AND HEALTH SCIENCES: PHENOTYPE DISTRIBUTIONS ACROSS GLEASON GROUPS

Task and frozen evidence. This case examines how Proliferation and Apoptosis phenotype scores differ across low, intermediate, and high Gleason groups and how the corresponding group centroids relate to the model’s phenotype directions. The frozen inputs contain 313 patient-specific Boolean models (23 low, 166 intermediate, and 124 high), the five PCA phenotype coordinates, and the explained variance of each component. The phenotype scores are semi-quantitative model probabilities, not direct measurements of cellular rates.

Reference evidence chain. The executable analysis preserves all patient rows, performs tie-corrected Kruskal–Wallis comparisons, and reconstructs both the PCA panel and the full group-wise score distributions shown in Figure 9. Proliferation differs across groups ($H = 12.36$, $p = 0.00207$), with a larger high-score tail in the high-Gleason group. Apoptosis also differs ($H = 25.55$, $p = 2.83 \times 10^{-6}$), but the source analysis states that it shows no clear grade-wise trend. The PCA centroid audit provides secondary visual context; none of these descriptive comparisons establishes a causal disease mechanism or treatment effect.

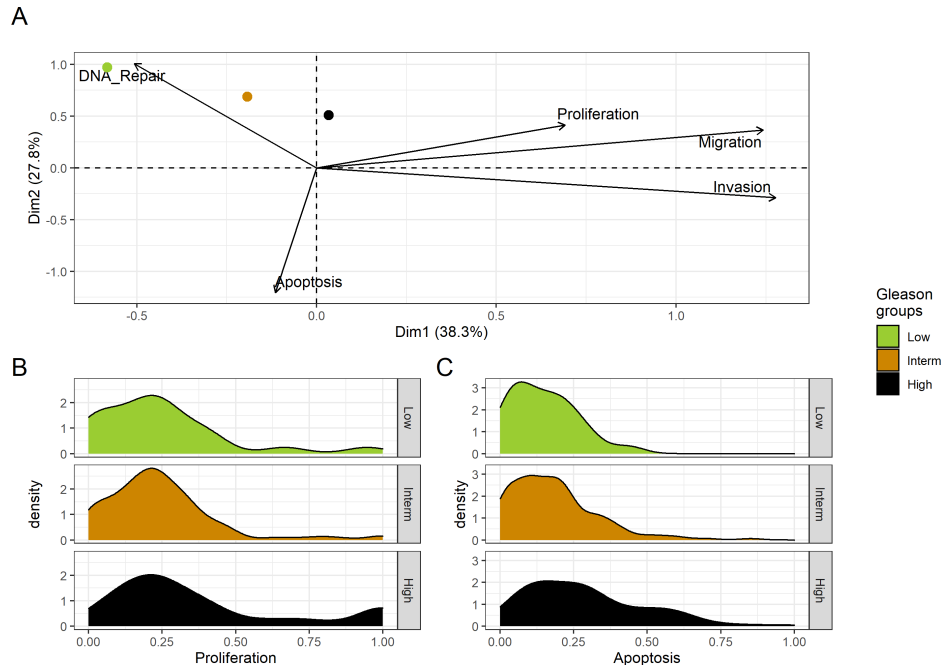


Figure 9: **Gleason phenotype case.** The frozen Figure 4 reference combines PCA phenotype directions and group centroids with complete Proliferation and Apoptosis distributions for the low, intermediate, and high Gleason groups. The two displayed dimensions explain 38.3% and 27.8% of the model variance.

F.6 PHYSICAL SCIENCES: POST-IMPACT FADING OF DIDYMOS

Task and frozen evidence. This case asks for Didymos’s post-impact fading rate, the time at which the fitted trend returns to the pre-impact absolute magnitude, and the associated projected dust velocity. One table contains all 27 plotted measurements and their observation-level uncertainties from 15 observers; a second identifies the author’s 21-observation fit subset. Visible reference constants specify the pre-impact magnitude, photometric aperture, pixel scale, and Earth distance.

Reference evidence chain. The executable analysis retains the author’s weighting (`sigma` set to squared magnitude error with `absolute_sigma=True`), fits the 21 selected observations, solves its crossing with the pre-impact magnitude, and converts the aperture geometry and elapsed time to velocity. The fitted observations and reference quantities are shown in Figure 10. The analysis recomputes a fading rate of $0.0579 \text{ magnitude day}^{-1}$, a crossing at 23.69 days, and a projected velocity of 0.366 m s^{-1} . The paper reports $23.7 \pm 0.7 \text{ days}$ and $0.37 \pm 0.01 \text{ m s}^{-1}$; the construction of those two published uncertainties is not fully exposed by the plotting tables and is therefore retained as source context rather than independently scored.

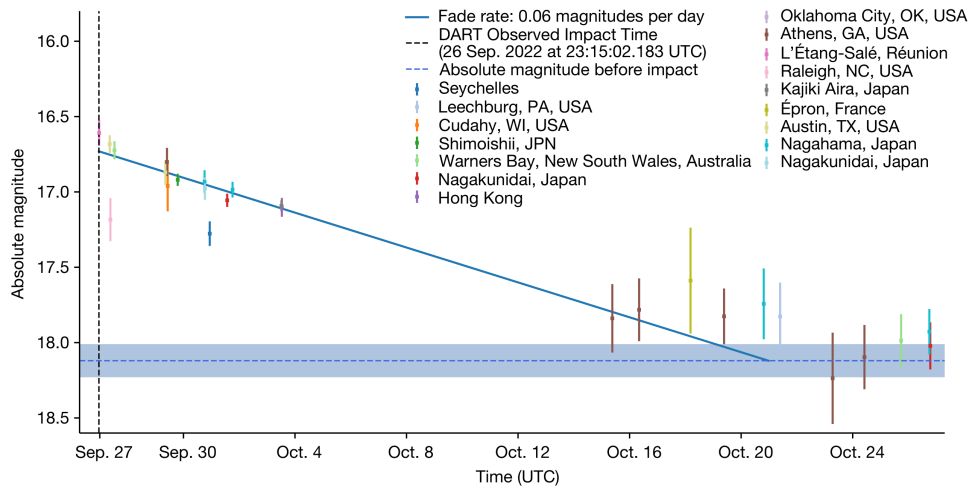


Figure 10: **Didymos post-impact fading case.** The frozen Figure 3 reference shows all 27 absolute-magnitude observations and error bars, the author-selected weighted fit, the impact-time marker, and the pre-impact magnitude with its uncertainty band. The astronomical magnitude axis is reversed, so upward movement denotes greater brightness.